

AI Scalping Trading Agent — System Report

Purpose: Paper-trading scalping agent for US stocks. Rule-based signals drive execution; LLMs can only **veto** or **rank** — they never initiate trades.

Mode: Alpaca paper account (<https://paper-api.alpaca.markets>)

Market: US equities only (not India/Reliance — Alpaca is US-only)

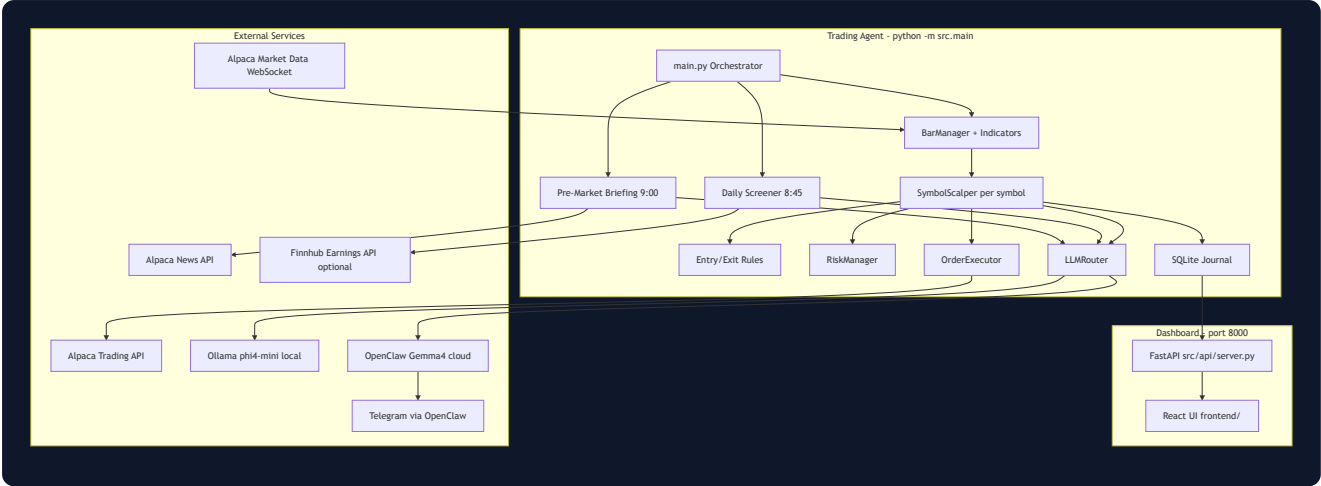
Session: Weekdays, **9:30–11:30 AM Eastern** (first 2 hours of regular session)

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1. High-Level Architecture

The system uses a **two-speed design**: fast Python rules on every 1-minute bar; slow LLM calls only for veto, ranking, and briefings.

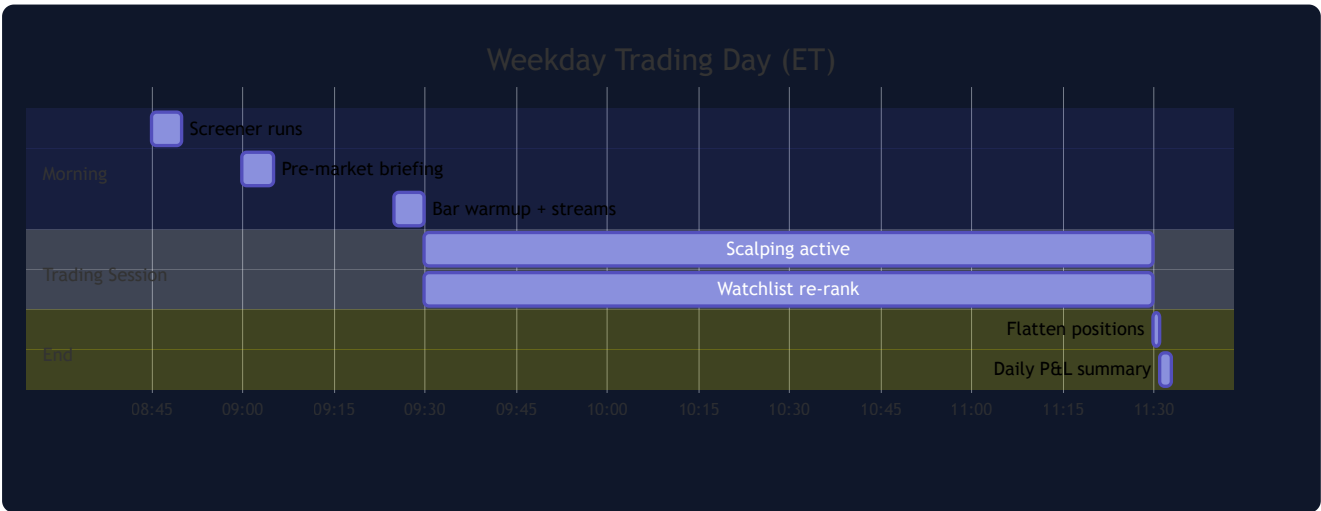


Design principles

Principle	Implementation
LLM never forces a trade	<code>trade_veto()</code> returns approve/reject only
Rules are fast	VWAP, RSI, volume evaluated on every bar
LLM is slow	3s timeout; fallback to OpenClaw if Ollama fails
Risk is hard-coded	Kill switch, position limits — not delegated to AI
Bracket orders protect crashes	Stop-loss + take-profit on Alpaca even if server dies

2. Daily Timeline (What Runs When)

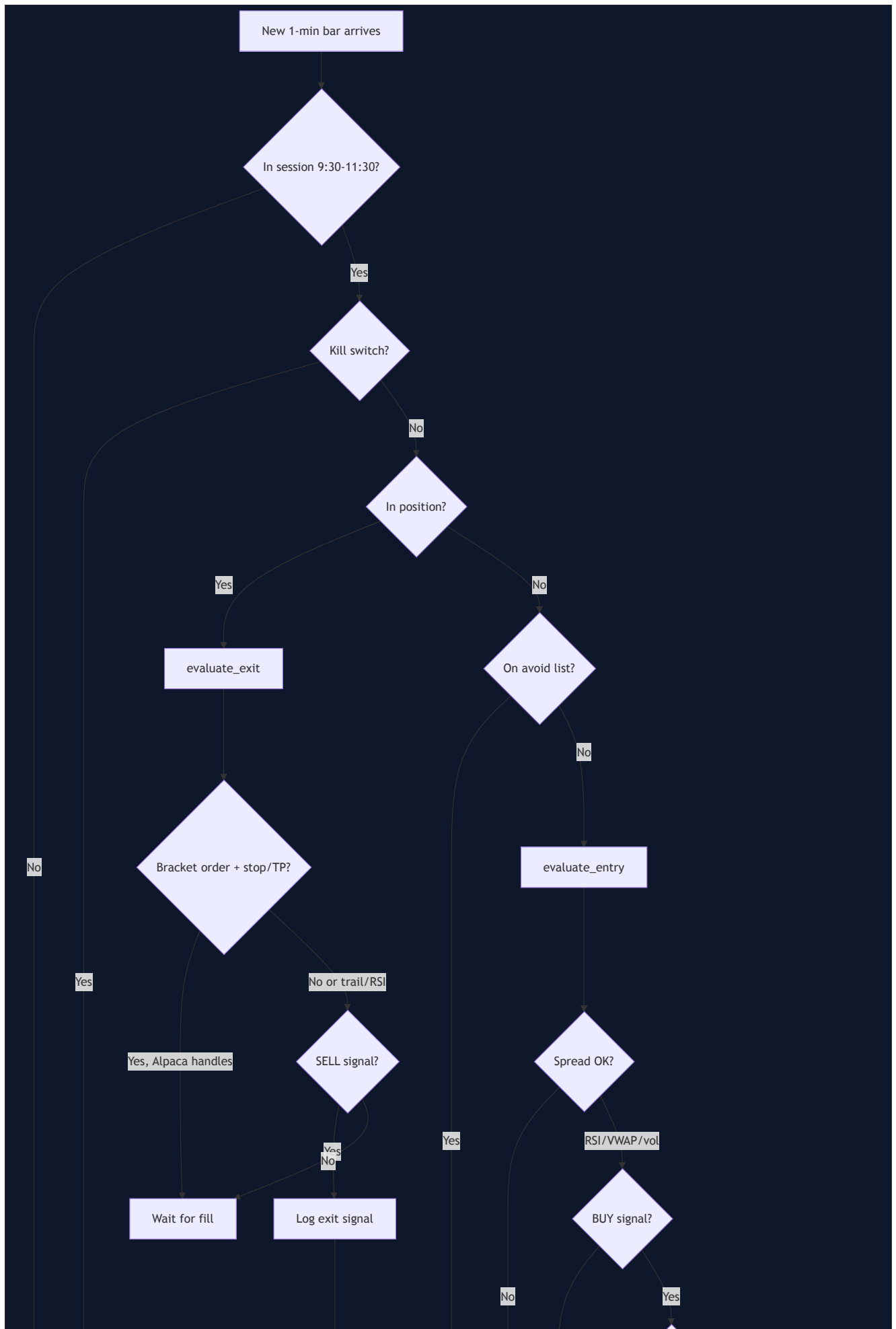
All times are **US Eastern** (America/New_York).



Time	Event	What happens
08:45	Daily screener	Fetches most-actives/movers → filters → enriches → LLM ranks 3 picks → watchlist = SPY + QQQ + 3 dynamic
09:00	Pre-market briefing	Fetches 24h news → keyword scan → LLM avoid/caution list → symbols blocked for entries
09:30	Session start	1-min bar stream live; scalper evaluates each symbol on every bar
Every 10 min	Watchlist re-rank	LLM re-prioritizes symbols (logged, informational)
11:30	Session end	All positions flattened; daily P&L saved; OpenClaw daily summary alert
Any time	Kill switch	If daily loss $\geq 2\%$ → flatten, stop agent, alert

3. Full Pipeline Diagram

Per-symbol trade decision flow (every 1-minute bar)





4. Configuration Reference (All Values)

4.1 Environment variables (`.env`)

Variable	Default / Example	Purpose
<code>ALPACA_API_KEY</code>	<i>(required)</i>	Alpaca paper API key
<code>ALPACA_SECRET_KEY</code>	<i>(required)</i>	Alpaca paper secret
<code>ALPACA_BASE_URL</code>	<code>https://paper-api.alpaca.markets</code>	Paper vs live endpoint
<code>ALPACA_DATA_FEED</code>	<code>iex</code>	<code>iex</code> (free) or <code>sip</code> (paid, faster)
<code>OLLAMA_HOST</code>	<code>http://127.0.0.1:11434</code>	Local Ollama server
<code>OLLAMA_MODEL</code>	<code>phi4-mini</code>	Primary LLM model
<code>OPENCLAW_GATEWAY_URL</code>	<code>http://127.0.0.1:18789</code>	OpenClaw gateway
<code>OPENCLAW_MODEL</code>	<code>gemma4:cloud</code>	Fallback cloud model
<code>OPENCLAW_ALERT_CHANNEL</code>	<code>telegram</code>	Alert destination
<code>ALERT_WEBHOOK_URL</code>	<i>(optional)</i>	Webhook if not using OpenClaw
<code>FINNHUB_API_KEY</code>	<i>(optional)</i>	Earnings calendar filter
<code>DASHBOARD_HOST</code>	<code>0.0.0.0</code>	Dashboard API bind address
<code>DASHBOARD_PORT</code>	<code>8000</code>	Dashboard port

4.2 Strategy (`config/settings.yaml` → `strategy`)

Setting	Value	Meaning
<code>rsi_period</code>	14	RSI lookback bars
<code>rsi_oversold</code>	35	Buy when $RSI \leq$ this (bounce setup)
<code>rsi_overbought</code>	65	Sell when $RSI \geq$ this (if in profit)
<code>volume_spike_ratio</code>	1.5	Current bar volume must be $\geq 1.5 \times$ 20-bar average
<code>vwap_deviation_pct</code>	0.15	Price must be $\geq 0.15\%$ below VWAP
<code>take_profit_pct</code>	0.20	Exit at $+0.20\%$ gain
<code>stop_loss_pct</code>	0.12	Exit at -0.12% loss
<code>trailing_stop_pct</code>	0.10	Exit if price drops 0.10% from session high (while profitable)

Entry condition (all must be true):

- $RSI \leq 35$

- Price $\leq$ VWAP – 0.15%
- Volume ratio $\geq$ 1.5
- Bid-ask spread < 0.03%

4.3 Session (session)

Setting	Value
timezone	America/New_York
start_time	09:30
end_time	11:30

4.4 Risk (risk)

Setting	Value	Meaning
max_risk_per_trade_pct	1.0%	Max equity risked per trade (drives position size)
daily_max_loss_pct	2.0%	Kill switch if equity drops 2% from session start
max_open_positions	2	Max simultaneous holdings
pdt_equity_threshold	\$25,000	Below this, PDT rules apply
max_day_trades	3	Max round-trips per day if equity < \$25k

Position size formula:

```

risk_amount = equity × 1%
stop_distance = entry_price × 0.12%
shares = floor(risk_amount / stop_distance)
capped at 10% of equity / price
minimum 1 share

```

4.5 LLM (llm)

Setting	Value	Meaning
enabled	true	Master switch for veto + ranking
confidence_threshold	0.7	Below this → fallback to OpenClaw
timeout_seconds	3	Ollama call timeout
watchlist_interval_minutes	10	Intraday re-rank frequency

4.6 Execution (`execution`)

Setting	Value	Meaning
<code>max_spread_pct</code>	0.03	Block entry if spread $\geq$ 0.03%
<code>use_bracket_orders</code>	true	Attach TP/SL to Alpaca on buy

Bracket prices (example @ \$100 entry):

- Take profit: \$100.20 (+0.20%)
- Stop loss: \$99.88 (−0.12%)

4.7 Briefing (`briefing`)

Setting	Value
<code>enabled</code>	true
<code>time</code>	09:00
<code>news_lookback_hours</code>	24
<code>news_limit</code>	10 articles

Keyword auto-flag: earnings, fda, guidance, downgrade, lawsuit, sec, investigation

4.8 Journal context (`journal_context`)

Setting	Value	Meaning
<code>enabled</code>	true	Use past trade stats in veto
<code>rsi_tolerance</code>	5	Match similar setups $\pm$ 5 RSI
<code>lookback_days</code>	30	History window
<code>min_trades_for_veto</code>	5	Need $\geq$ 5 similar past trades
<code>min_win_rate</code>	0.4	Block if win rate < 40%

4.9 Screener (screener)

Setting	Value	Meaning
enabled	true	Dynamic watchlist
mode	hybrid	hybrid dynamic static
anchor_symbols	SPY, QQQ	Always on watchlist
dynamic_slots	3	LLM-picked symbols per day
run_time	08:45	When screener runs
candidate_pool_size	30	Top N from Alpaca screener
min_price	\$15	Filter cheap stocks
max_price	\$500	Filter very expensive
min_volume	1,000,000	Min daily volume
max_percent_change	8.0%	Skip extreme movers
finnhub_enabled	true	Earnings-day filter (needs API key)
exclude_yesterday_losers	true	Skip symbols that lost money yesterday

Today's watchlist (hybrid mode): SPY, QQQ + 3 screener picks = **5 symbols**

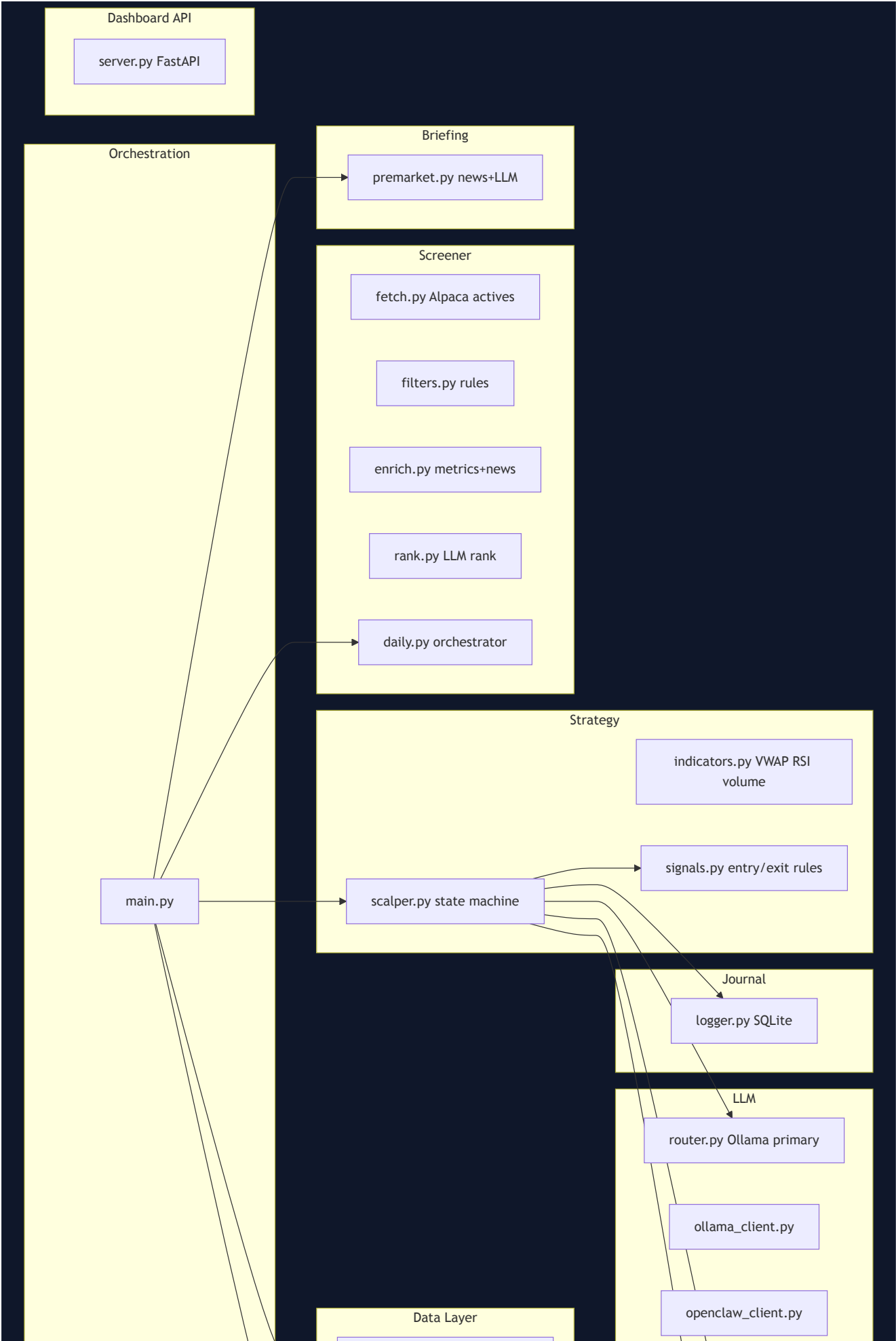
4.10 Journal database (journal)

Setting	Value
db_path	data/trades.db

4.11 Static symbols (fallback when screener.mode: static)

AAPL, MSFT, SPY, QQQ, NVDA

5. Module-by-Module Breakdown





Module	File	Responsibility
Orchestrator	<code>src/main.py</code>	Startup, screener, briefing, streams, session guard, kill switch
Quote/Bar stream	<code>src/data/stream.py</code>	Alpaca WebSocket: live quotes + 1-min bars
Bar warmup	<code>src/data/bars.py</code>	Load 2 days of 1-min history so RSI/VWAP ready at open
Order fills	<code>src/data/trade_stream.py</code>	Alpaca trading WebSocket for fill events
Indicators	<code>src/strategy/indicators.py</code>	VWAP, RSI(14), 20-bar avg volume, spread from quote
Signals	<code>src/strategy/signals.py</code>	Pure rule evaluation — no AI
Scalper	<code>src/strategy/scalper.py</code>	Per-symbol state machine: IDLE → BUY → IN_POSITION → SELL
Orders	<code>src/execution/orders.py</code>	Market/limit/bracket orders via Alpaca
Positions	<code>src/execution/positions.py</code>	Flatten all at session end
Risk	<code>src/risk/manager.py</code>	Kill switch, max positions, PDT guard
Sizing	<code>src/risk/sizing.py</code>	Shares from 1% risk / 0.12% stop
LLM router	<code>src/llm/router.py</code>	Ollama first, OpenClaw fallback, fail-safe reject
Screener	<code>src/screener/*</code>	Daily stock picking pipeline
Briefing	<code>src/briefing/premarket.py</code>	News risk before open
Journal	<code>src/journal/logger.py</code>	All signals, trades, events, watchlist → SQLite
Dashboard API	<code>src/api/server.py</code>	REST API for React UI
Frontend	<code>frontend/</code>	React dashboard on port 8000

6. Entry Signal Logic (Buy)



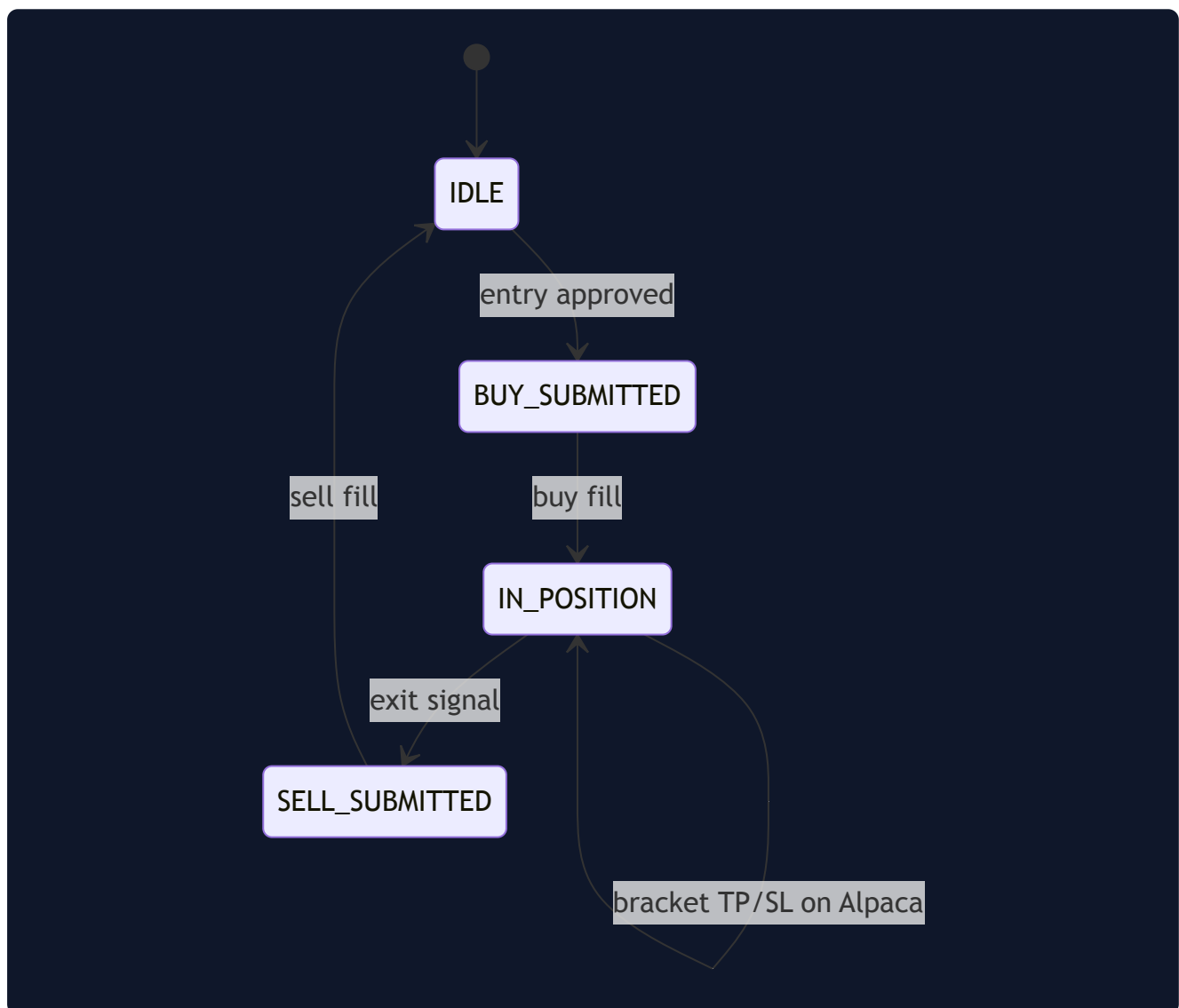
Syntax error in text
mermaid version 11.15.0

Logged to `signals` table: `signal_type=entry` , `details=vwap_rsi_volume_bounce` , plus RSI/VWAP/volume and LLM reason.

If vetoed: `signal_type=entry_vetoed` with LLM or journal reason.

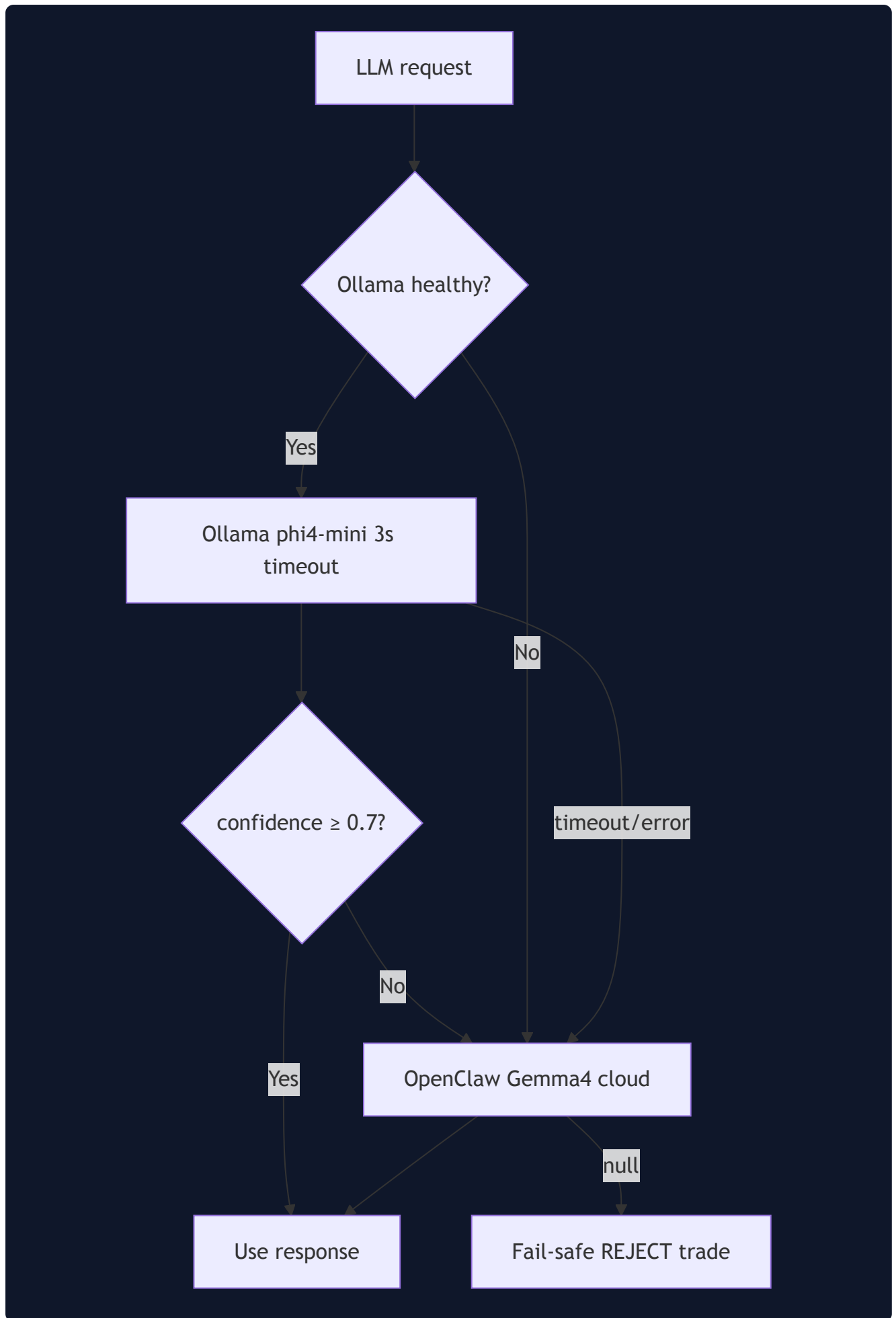
7. Exit Signal Logic (Sell)

Reason	Condition	Who executes
<code>stop_loss</code>	$P\&L \leq -0.12\%$	Alpaca bracket (if enabled) or bot
<code>take_profit</code>	$P\&L \geq +0.20\%$	Alpaca bracket (if enabled) or bot
<code>trailing_stop</code>	Down 0.10% from high since entry, still profitable	Bot (market sell)
<code>rsi_overbought</code>	$RSI \geq 65$ and $P\&L > 0$	Bot (market sell)



Logged: `signals` table (`exit` + reason), `trades` table (sell with P&L and exit reason).

8. LLM Layer (Ollama + OpenClaw)



LLM tasks (what AI does)

Task	When	Model	Can trade?
Trade veto	Each buy signal	Ollama → OpenClaw	Approve/reject only
Screener rank	8:45 AM	Ollama → OpenClaw	Pick 3 symbols
Pre-market briefing	9:00 AM	Ollama → OpenClaw	Avoid/caution lists
Watchlist rank	Every 10 min	Ollama → OpenClaw	Informational only
Daily summary	11:30 AM	OpenClaw	Alert text only
Trade alerts	On fill	OpenClaw → Telegram	Notification only

LLM cannot:

- Pick random stocks outside screener rules
 - Force a buy without a rule signal
 - Override kill switch or risk limits
 - Change stop-loss / take-profit percentages
-

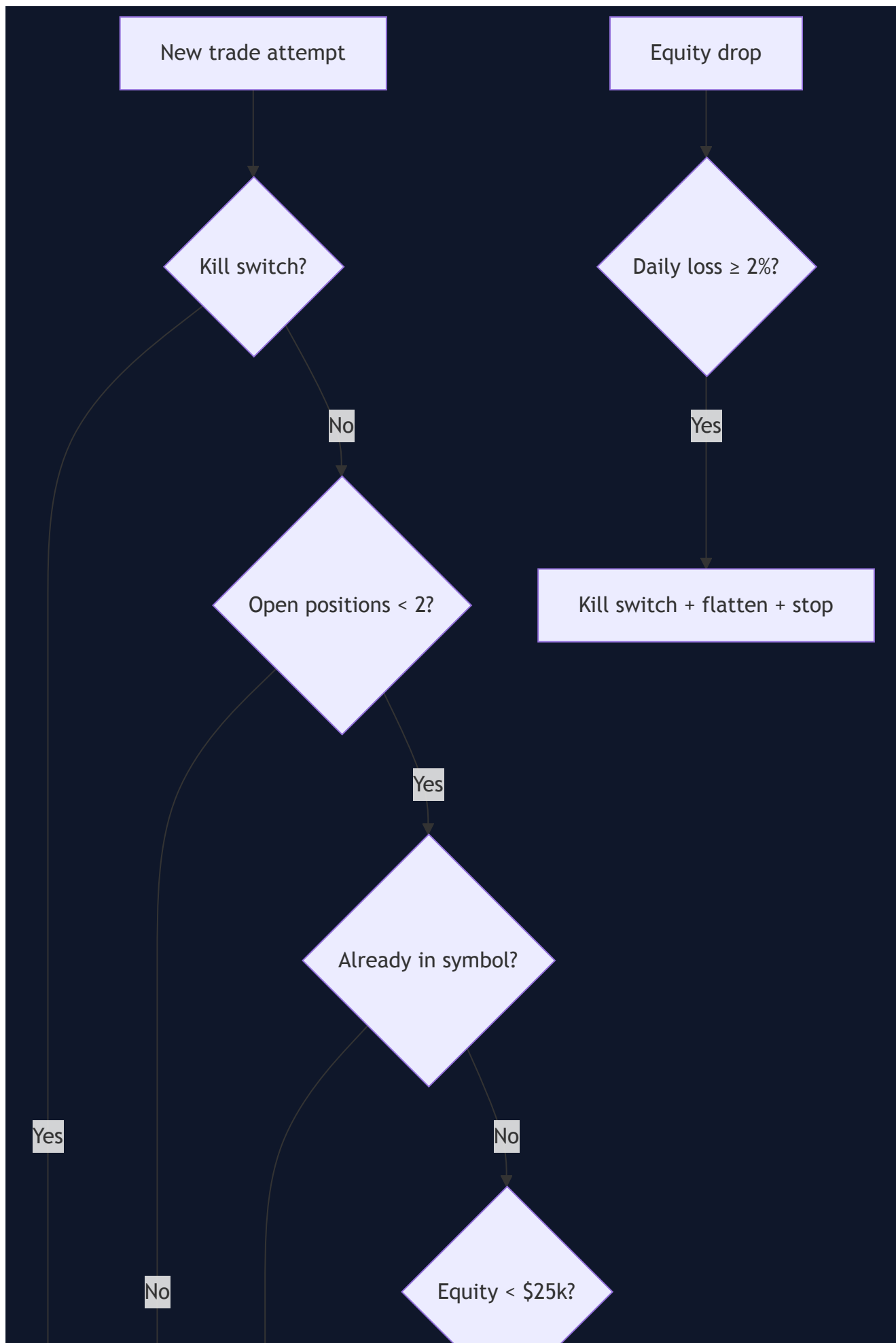
9. Risk Management

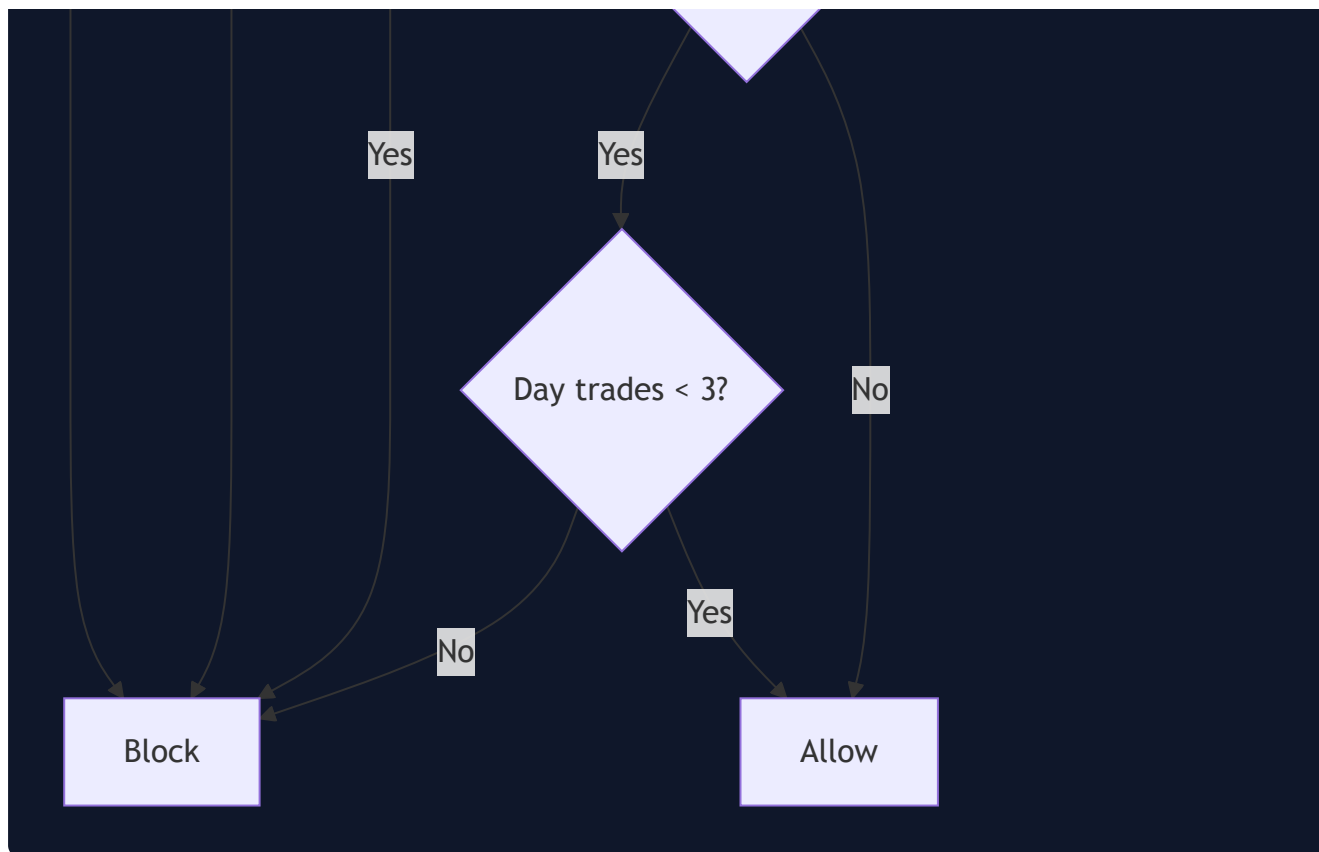
The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

The second part of the paper focuses on the methodology used in the study. It describes the process of selecting participants, collecting data, and analyzing the results. The authors emphasize the importance of using a mixed-methods approach to gain a comprehensive understanding of the research topic.

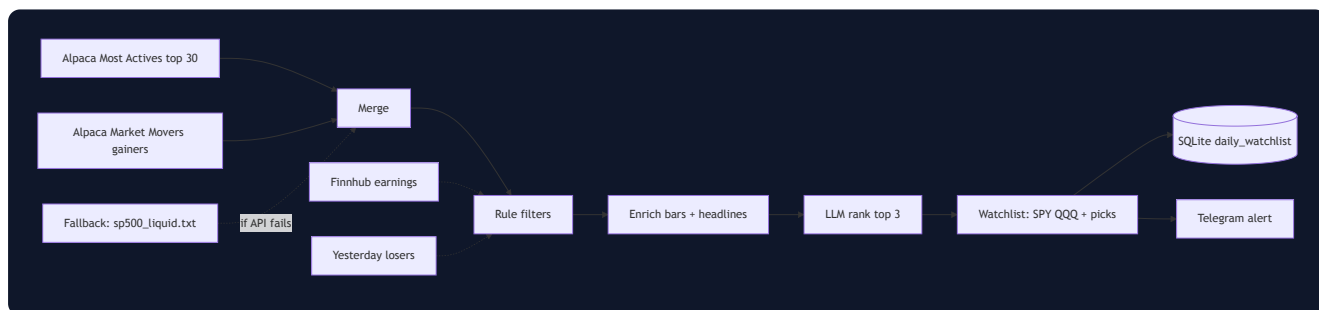
The third part of the paper presents the findings of the study. It discusses the results of the quantitative data analysis and the insights gained from the qualitative interviews. The authors conclude that there are significant differences in learning outcomes between the two groups, and these differences can be attributed to cultural factors.

The final part of the paper discusses the implications of the findings for future research and practice. It suggests that educators should be aware of the cultural context of their students and tailor their teaching strategies accordingly. The authors also recommend further research to explore the underlying mechanisms of the cultural differences.





10. Daily Screener Pipeline



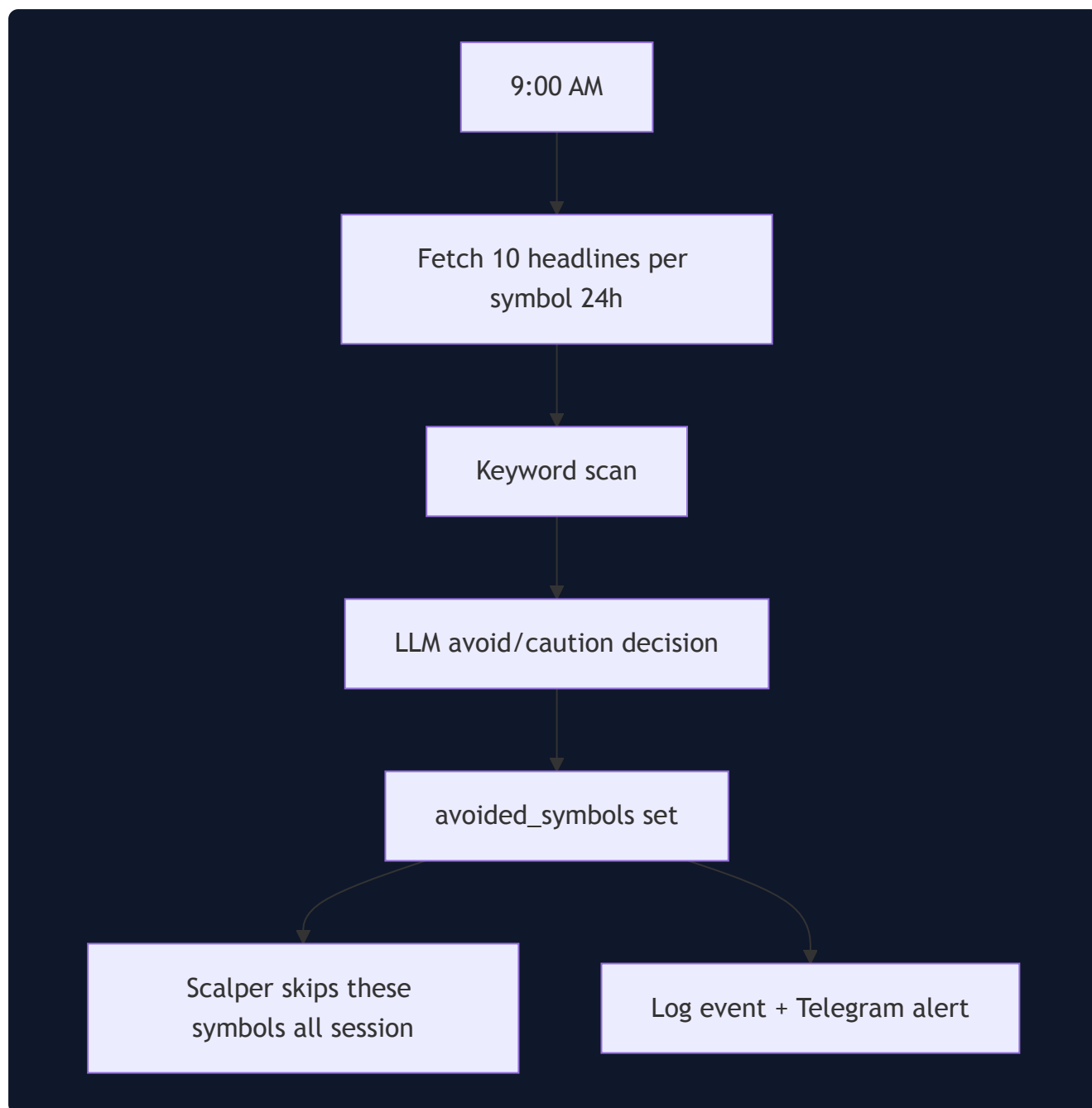
Filters applied

1. Price \$15 – \$500
2. Volume $\geq 1\text{M}$
3. $|\% \text{ change}| \leq 8\%$
4. Not on earnings today (Finnhub)
5. Not a yesterday loser (from journal)
6. Anchors (SPY/QQQ) excluded from dynamic pool

Enrichment per candidate

- Gap %, volume ratio, ATR %, recent headlines (Alpaca News)

11. Pre-Market Briefing Pipeline



Blocked symbols: No new entries for rest of session (existing positions still managed).

12. Order Execution

Order type	When used
Bracket market BUY	Default entry — includes TP limit + SL stop on Alpaca
Market BUY	If <code>use_bracket_orders: false</code>
Limit SELL	Stop-loss / take-profit exits (non-bracket)
Market SELL	Trailing stop, RSI overbought
Cancel open orders	Before software-initiated sell

13. Data Storage (SQLite)

Database: `data/trades.db`

events		
int	id	PK
text	ts	
text	event_type	
text	message	

same session

signals		
int	id	PK
text	ts	
text	symbol	
text	signal_type	
text	details	
text	llm_action	
real	llm_confidence	
text	llm_reason	
real	rsi	
real	vwap_dev	
real	volume_ratio	

leads to

daily_watchlist		
text	date	PK
text	symbol	PK
text	source	
int	rank	
text	metrics_json	
text	reason	

symbol on date

daily_pnl		
text	date	PK
real	starting_equity	
real	ending_equity	
real	net_pnl	
int	trade_count	
int	win_count	

aggregates

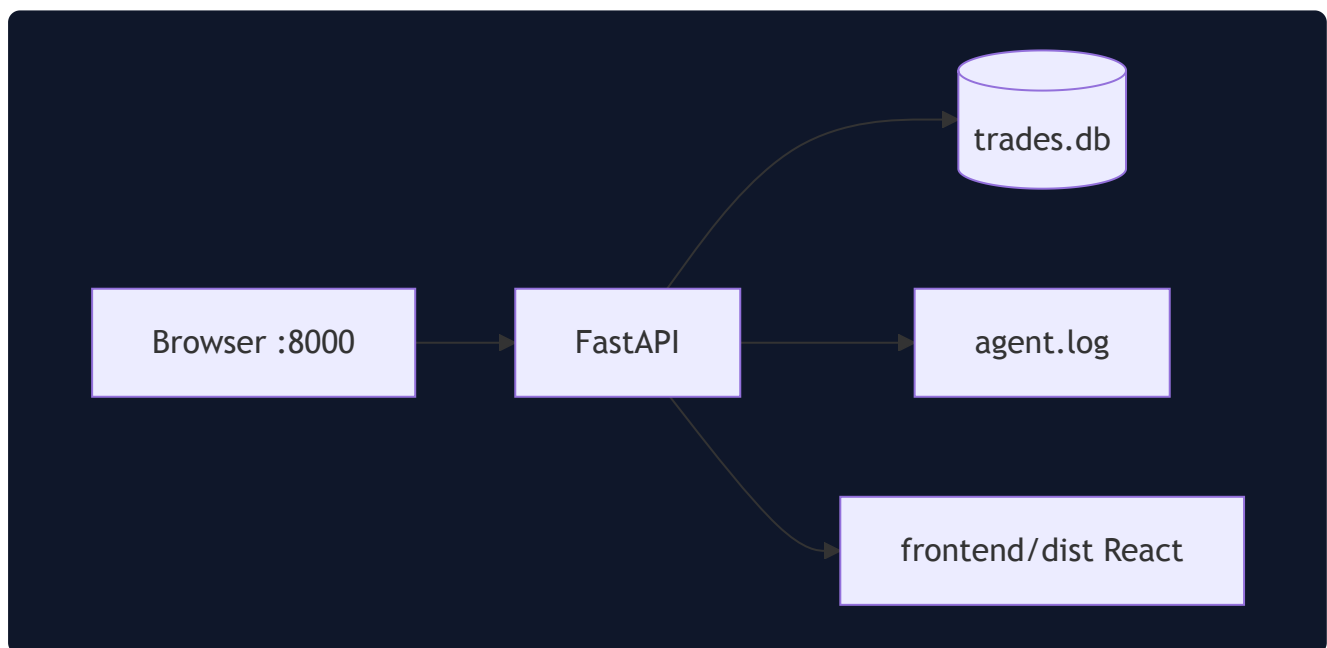
trades		
int	id	PK
text	ts	
text	symbol	
text	side	
real	qty	
real	price	
text	order_id	
real	pnl	
text	reason	

What gets logged

Event	Table	Example
Buy signal approved	signals	entry + LLM reason
Buy vetoed	signals	entry_vetoed
Exit triggered	signals	exit, stop_loss
Order filled	trades	buy/sell + P&L
Pre-market briefing	events	premarket_briefing
Watchlist re-rank	events	watchlist_rank
Screener result	daily_watchlist	per symbol + reason
End of day	daily_pnl	equity + win count

Agent file log: `logs/agent.log` (all Python logging)

14. Dashboard (React Frontend)



Page	Data source	Shows
Overview	summary + stats + watchlist	P&L, win rate, config
Trades & Reasons	round-trips	Why bought, why sold, AI reason
AI Decisions	signals	All entries, vetoes, exits
Screener	daily_watchlist	Daily picks + metrics
Events	events	Briefings, re-ranks
P&L History	daily_pnl	Chart + table
Agent Logs	agent.log	Live tail, 5s refresh

Run: `python scripts/run_dashboard.py` → <http://localhost:8000>

15. Alerts & Notifications

All alerts go through **OpenClaw** → **Telegram** (configurable).

Alert	Trigger
Agent Started	<code>python -m src.main startup</code>
Daily Screener	8:45 watchlist + reasons
Pre-Market Briefing	Avoid/caution lists
Trade Fill	Buy or sell fill + P&L
Daily P&L Summary	11:30 session end
Kill Switch	Daily max loss hit

16. How to Run Everything

```
# 1. Setup (once)
cd AI_Stock_portfolio
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
cp .env.example .env    # add Alpaca keys

# 2. Start Ollama (Kali/Mac)
ollama serve
ollama pull phi4-mini

# 3. Start OpenClaw gateway (for fallback + alerts)

# 4. Trading agent (weekdays, waits for 8:45 if started early)
python -m src.main

# 5. Dashboard (anytime, read-only)
cd frontend && npm install && npm run build && cd ..
python scripts/run_dashboard.py
# → http://localhost:8000

# 6. Reports (CLI)
python scripts/paper_report.py
python scripts/screener_report.py
python scripts/backtest.py
```

Systemd (Kali Linux)

Service	File	Command
Trading agent	systemd/trading-agent.service	python -m src.main
Dashboard	systemd/trading-dashboard.service	python scripts/run_dashboard.py

17. File Map

```
AI_Stock_portfolio/
├─ config/settings.yaml      # All tunable parameters
├─ .env                      # API keys (not committed)
├─ data/
│   ├── trades.db            # SQLite journal
│   └─ universe/sp500_liquid.txt # Screener fallback universe
├─ logs/agent.log            # Runtime log file
├─ scripts/
│   ├── run_dashboard.py     # Start dashboard API
│   ├── paper_report.py      # CLI daily report
│   ├── screener_report.py   # CLI screener report
│   └─ backtest.py           # Historical backtest
├─ src/
│   ├── main.py              # Agent entry point
│   ├── config.py            # Config loader
│   ├── api/server.py        # Dashboard REST API
│   ├── data/                # Streams + bars
│   ├── strategy/            # Indicators, signals, scalper
│   ├── execution/           # Orders, positions
│   ├── risk/                # Risk manager, sizing
│   ├── llm/                 # Ollama, OpenClaw, prompts
│   ├── screener/            # Daily stock picker
│   ├── briefing/            # Pre-market news
│   └─ journal/              # SQLite logger
├─ frontend/                 # React dashboard
└─ systemd/                  # Linux service units
```

Quick Reference Card

Question	Answer
What stocks does it trade?	SPY + QQQ + 3 daily screener picks (hybrid mode)
When does it trade?	9:30–11:30 AM ET, weekdays
What triggers a buy?	RSI≤35 + below VWAP + volume spike + tight spread
What triggers a sell?	−0.12% stop, +0.20% target, 0.10% trail, RSI≥65
What does AI do?	Veto bad entries, rank screener, briefing, alerts
What does AI NOT do?	Pick trades alone, override risk, go live without rules
Where is history?	data/trades.db + dashboard at :8000
Paper or live?	Paper by default (ALPACA_BASE_URL)

Generated from codebase state. Edit `config/settings.yaml` and `.env` to change behavior.